

	Credit Hours System		Program:	Biomedical	
	Academic Year:	2024	Semester:	Spring	
	Course Code:	SBES160	Course Title:	Computer Vision	
	Day:	Wednesday	Date:	May 1 st , 2024	
	Time:	1 Hr	Full Mark:	22	

1. The 2D-DFT combines frequencies; i.e. instead of representing two peaks for the signal in 1D-DFT, it combines them. State and explain whether true or false (2 pt.)

2. The two images (im1 to the left: Nietzsche, and im2 to the right: kitten) below are in grayscale and normalized to be in the range of [0, 1], thus: (10 pt.)



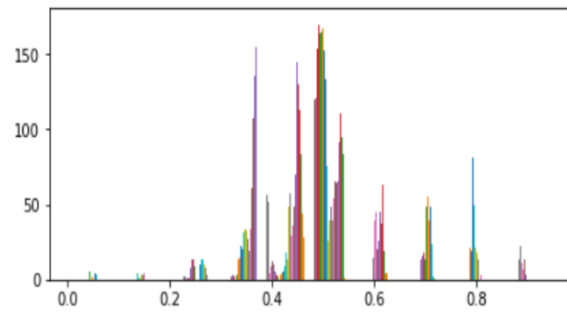
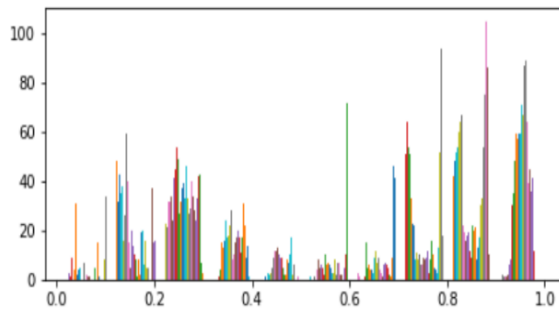
- i) The contrast of image1 is ----- than that of the kitten: (2 pt.)

a) less	b) greater
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- ii) the high frequency content of image1 is ----- than that of the kitten's: (2 pt.)

a) less	b) greater
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- iii) the images below (left and right) represent the histogram for the Nietzsche and kitten; NOT RESPECTIVELY (i.e. NOT SORTED). Left image is the histogram of the ----- image: (2 pt.)

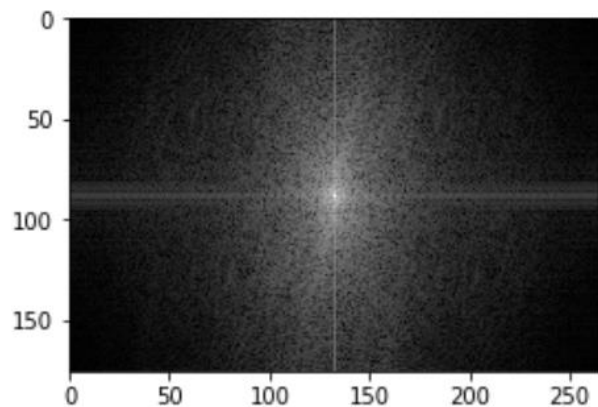
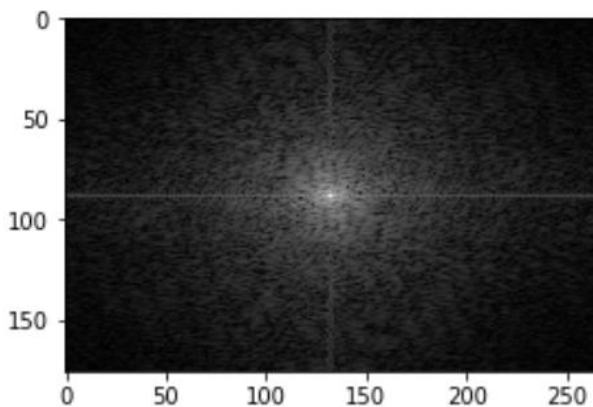


a)	Nietzsche	b)	kitten
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- iv) the images below (left and right) represent the Fourier transform magnitude for the Nietzsche and kitten; NOT RESPECTIVELY (i.e. NOT SORTED). Note: the two images below are centered and log-transformed.

Left image is the spectrum of the ----- image: (2 pt.)

a)	Nietzsche	b)	kitten
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- v) $F_{\text{nietzsche}}(0,0)$ is ----- than $F_{\text{kitten}}(0,0)$ (2 pt.)

a) Less	b) greater
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3. Apply a 3×3 median filter for the image below and write the output matrix. DO NOT apply PADDING. (3 pt.)

100	115	100	95	50
85	95	90	75	50
120	95	95	75	50

4. When you work on frequency domain; the two essential pre-processing steps are and Moreover, it is recommended to (3 pt.)
5. You are working on X-ray images in an orthopedic clinic. Please design an algorithm to detect bone fracture and differentiate between crack within the bone and complete dissociation between two bone parts(4 pt.)